Pavan Pandya

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Education

Indiana University - Bloomington **August 2023 – May 2025**
Master of Science in Computer Science | GPA: 3.3/4.0 *Bloomington, Indiana, USA*
Courses: Applied Algorithms, Elements of AI, Intro to Statistics, Applied Machine Learning, Software Engr. - 1, Engr. Cloud Computing

LDRP Institute of Technology and Research **August 2019 – June 2023**
Bachelors of Engineering in Computer Engineering | GPA: 3.82/4.0 *Gandhinagar, Gujarat, India*
Courses: Data Structure & Algorithms, Object-Oriented Programming, Database Management System, Machine Learning, Neural Networks & Deep Learning, Natural Language Processing, Image Processing, Optimization Techniques, Computer Networks

Technical Skills

Programming Languages: Python, R, C, SQL
Frameworks: PyTorch, TensorFlow, Keras, OpenCV, Langchain, Flask, Django, FastAPI
Libraries: Scikit-Learn, Pandas, NumPy, NLTK, SciPy, Hugging Face, spaCy, Regex, Matplotlib, Seaborn, Selenium
Database Langauge: PostgreSQL, MySQL, MongoDB
Web Technologies: HTML, CSS, JavaScript, NodeJs, ExpressJs, RestAPI, Bootstrap, SweetAlert
Tools: Git, Github, Jira, Linux, AWS, GCP, Shell, Docker, Tableau, Postman, MS Excel, MLflow, Dagshub, VS Code, Google Colab

Experience

Hate Speech and Offensive Content Identification | Link **June 2021 – Present**
Part-Time Student Organizer *Gandhinagar, Gujarat, India*

- Led backend development, creating a robust API for hate speech identification. Rigorously tested with Postman and documented for smooth integration, ensuring system efficiency and maintainability.
- Engineered a novel user registration system in MongoDB, contributing to the core functionality of the annotation platform. Demonstrated strong teamwork and communication through active participation in meetings.
- Elevated to Student Organizer, led the identification of hate speech in Gujarati tweets. Spearheaded the strategic collection of 1200 tweets using Selenium and actively participated in model fine-tuning.
- Played a pivotal role in hands-on model fine-tuning through manual annotation of 200 tweets, leveraging Hugging Face to achieve 10% improvement in performance. further contributing to 7% reduction in false positives

Dhirubhai Ambani Institute of Technology **Nov. 2022 – June 2023**
Research Intern *Gandhinagar, Gujarat, India*

- Gathered >1.5 million tweets through Twitter API & Selenium, designed an efficient MongoDB schema for streamlined analysis.
- Implemented advanced text preprocessing techniques and harnessed BERT embeddings to accurately capture nuanced sentiment, aligning with industry-leading research practices, resulting in a 20% improvement in sentiment analysis accuracy.
- Extracted inflation-related tweets using BERTopic, increased model accuracy to 77% with few-shot learning on 300 manually annotated tweets for enhanced sentiment analysis.
- Collaboratively refined model parameters and provided key insights into public inflation perception, demonstrating strong teamwork and communication skills.

Projects

Kidney Disease Classification | *Python, Deep Learning, MLflow, DVC, AWS, Docker, GitHub Actions* **Nov. 2023 – Jan. 2024**

- Achieved a 15% improvement in kidney disease classification accuracy by developing and fine-tuning a deep learning model. Utilized MLflow for experiment tracking, enabling precise identification of model enhancements.
- Implemented a streamlined deployment process that reduced deployment time by 30%. Created an IAM user, built a Docker image, pushed it to AWS ECR, launched an EC2 instance, and deployed the image using GitHub Actions, resulting in an efficient and automated AWS CI/CD pipeline.

Uber Data Analysis Pipeline using GCP | *Python, Data Extraction, Data transformation, GCP, Mage AI, Looker* **Aug. 2023 – Sept. 2023**

- Orchestrated the development of a robust GCP data pipeline using Python, BigQuery, Looker, and Mage AI to analyze NYC taxi trip data, resulting in a 20% improvement in data processing efficiency.
- Enhanced decision-making capabilities by uncovering key patterns and trends in the BigQuery data, translating findings into insightful visual dashboards with Looker. This initiative led to a 15% increase in operational efficiency.

AgriOracle - IBM Hack Challenge 2021 Winner | *Python, IBM Watson Studio and Chat-bot* **June 2021 – Aug. 2021**

- Led the development of an AI-driven application, utilizing soil composition and weather data to achieve a 20% increase in crop yield and a 15% reduction in water usage for optimal crop selection.
- Implemented season-specific crop yield projections and price forecasts, achieving an 18% increase in accurate yield predictions and a 25% improvement in price forecasting accuracy over a 12-month horizon.